KRATIKA RATHI

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science (Artificial Intelligence) — *December 2027*
Relevant Coursework: Theoretical Machine Learning (PhD-level), Algorithm Design and Analysis, Natural Language Processing

The University of British Columbia — Vancouver, BC
Bachelor of Science in Statistics — *May 2025*
Relevant Coursework: Bayesian Statistics, Multivariate Statistical Analysis, Probability, Time Series and Forecasting

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, JavaScript
Machine Learning & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, statsmodels
LLM & NLP: Hugging Face Transformers, LangChain, Ollama, OpenAI API, Anthropic API, SAELens
Tools & Visualization: Git, Linux, Docker, Tableau, Power BI

EXPERIENCE

Graduate Researcher — June 2026 - October 2026
Melady Lab, University of Southern California — *Los Angeles, CA*
- Designed STRATA, an LLM-agent framework that decomposes agent skills into reusable procedure cores and environment-specific adapters to improve cross-environment skill transfer.
- Developed self-improving agents that revise written instructions from execution feedback, with safeguards that detect regressions in previously explored environments.
- Curated benchmark datasets and built evaluation pipelines measuring task success, transfer performance, and generalization across environments.

Actuarial Intern — January 2024 - August 2024
iA Financial — *Vancouver, BC*
- Performed monthly reserve calculations for the Special Markets Group using chain-ladder methods and analyzed reserve adequacy and portfolio performance.
- Conducted profitability studies and applied GARCH modeling to forecast claim patterns, contributing to a 15% improvement in financial forecasting.
- Designed and validated survey instruments that reduced data-entry and validation errors by 10%; built pricing, reserving, and valuation models to support risk assessment.

Data Analyst — January 2023 - April 2023
Orennia — *Calgary, AB*
- Built and maintained multinomial logistic regression models predicting power-plant viability from capacity, location, and size to support investment decisions across U.S. energy markets.
- Automated forecast generation with reusable Python scripts, improving the efficiency and reproducibility of recurring modeling workflows.

Data Analyst — May 2022 - December 2022
Animal Welfare Program, University of British Columbia — *Vancouver, BC*
- Applied mixed-effects models to assess the repeatability and predictability of cattle feeding and drinking behavior for animal-welfare monitoring.
- Analyzed high-volume behavioral data to identify dominance hierarchies and support early detection of illness, lameness, and missing animals.
- Optimized Python data pipelines and contributed to statistical analysis, visualization, interpretation, and technical research reporting.

RESEARCH

Classifying Mantle Compositions Using Geochemical Data from Hawaiian Lavas — *Manuscript in Preparation*
- Built an end-to-end machine-learning pipeline for trace-element geochemical data, including missing-value imputation, feature engineering, and supervised model evaluation.
- Trained and compared logistic regression, support vector machine, random forest, and gradient boosting classifiers to identify discriminative signatures of Hawaiian mantle compositions.
- Assessed trace-element predictors as lower-cost alternatives to isotope-based measurements, connecting model outputs to the composition and evolution of Earth's mantle.

PROJECTS

Where Does the Truth Emerge? Activation Patching in TriviaQA — *In Progress*
- Built an evaluation pipeline over 2,000 TriviaQA questions and 12,000 prompt variants using Qwen3.5-0.8B and Gemma3-1B to study wording-dependent factual errors.
- Applied layerwise logit-lens analysis and bidirectional residual-stream activation patching, with self-patch and unrelated-donor controls, to repair incorrect answers and reverse correct ones.
- Identified peak causal layers at Qwen layer 23 and Gemma3-1B layer 16; achieved wrong-to-right flips for 199/202 and 243/263 questions, respectively, with Wilcoxon signed-rank $p < 10^{-16}$.

Fault in Our Stars: Astronomical Light-Curve Classification | GitHub
- Built an end-to-end Python and scikit-learn time-series classification pipeline over 15,523 processed Kepler light curves, distinguishing confirmed exoplanets, eclipsing binaries, variable stars, and false positives.
- Implemented robust preprocessing with MAD-based sigma clipping, median normalization and detrending, KIC-level deduplication, and interpolation of variable-length flux sequences to 300-point feature vectors.
- Trained a 300-tree, class-balanced Random Forest with stratified splitting and SMOTE; evaluated it using macro-F1, per-class precision/recall, and confusion matrices, establishing a 0.57 macro-F1 four-class baseline.